

Problem Set 7

Instructor: Robert Rand

1 Permutations, Combinations, Sundaes and Identities

Problem 1: For each of the following questions, answer how many possible passphrases you can obtain. Justify your answer. (Do not reduce the answer to a concrete number, but simplify expressions where possible.)

1. You select 6 words at random from the 7776 word Diceware wordlist (<https://theworld.com/~reinhold/diceware.html>), where repetition is possible. (For instance, “bravo” can appear twice in the phrase.)
2. You select 6 *different* words from the list. (For example, you cross out a word when it’s selected.)
3. You select 6 different words and sort them in lexicographic order.

Bonus (Ungraded) Develop a variant on any of the schemes above with as many or more possible phrases than any of the preceding ones. Determine how many phrases are possible and then use the scheme in practice. (Don’t submit the generated phrase.)

Problem 2: Baskin-Robbins has 31 flavors of ice cream. Are there more ways of choosing 5 flavors for your ice cream sundae or 28 flavors? Justify your answer without using a calculator or computing the precise numbers by hand.

Problem 3: How many total ways are there of creating Baskin-Robbins sundaes, using any number (from 1 to 31) of flavors? Simplify your expression as much as possible. (No number crunching; quantity and order of flavors don’t matter.)

Problem 4: We have a standard deck of 52 cards plus one joker. How many ways are there to choose a five card hand that includes the joker? What about about a hand that excludes the joker? What do I get if I combine the terms (simplified but not computed)?